

Hotel Bar Inventory Forecasting & Par Level Recommendation System

1. Business Problem & Operational Impact

The hotel chain operates multiple bars where inventory availability must be balanced against excess stock. **Stockouts** of popular brands can affect guest experience, reduce sales opportunities, and interrupt bar operations. **Overstocking** slow-moving products ties up working capital, occupies storage space, and can increase shrinkage or wastage.

The objective is to transform transaction-level bottle balance records into daily consumption time series, forecast demand for each bar-brand combination, recommend inventory par levels, and evaluate the replenishment policy through historical backtesting.

2. Data Preparation & Assumptions

The dataset contains opening balance, purchases, consumption, closing balance, timestamp, bar, alcohol type, and brand. It covers **6 bars and 16 brands** from January 2023 to January 2024. Timestamps were converted to datetime and records were aggregated to daily consumption by bar and brand.

The inventory conservation relationship, **Closing = Opening + Purchase – Consumed**, was checked. Most records matched closely; small discrepancies in a limited number of rows were retained as possible measurement or rounding differences.

A complete Date × Bar × Brand grid was created, with missing combinations represented as zero consumption. This is important because **83.98%** of daily bar-brand records have zero consumption.

Assumptions: supplier lead time = 3 days; target service level = 95% ($Z = 1.645$); missing combinations represent zero recorded consumption; the backtest uses actual test-period demand; supplier capacity, minimum order quantities, and variable delivery times are not modeled.

3. Demand Forecasting Approach

A chronological train-test split was used to prevent future information from entering model training. Features include previous-day, 7-day and 14-day consumption, previous 7-day average consumption, day of week, bar, and brand.

A **Random Forest Regressor** was selected as the predictive model because it can capture nonlinear relationships among lagged demand, calendar information, bar, and brand without requiring a strong linearity assumption. A simple **7-day lag baseline** was also implemented to provide a practical benchmark.

Rolling features use only previous observations to avoid data leakage. The high frequency of zero-demand observations makes this an intermittent-demand forecasting problem.

4. Model Performance & Failure Modes

Metric	Random Forest	7-Day Lag Baseline
Overall MAE	98.42 ml	97.37 ml
Overall WAPE	176.85%	174.98%
MAE on demand days	281.97 ml	320.37 ml

The 7-day lag baseline performed slightly better on overall test MAE. However, on days where actual consumption occurred, the Random Forest had lower MAE. This illustrates the difficulty of forecasting highly intermittent demand. Positive predictions on some zero-demand days and high WAPE are important failure modes. Intermittent-demand-specific methods could improve the approach.

5. Dynamic Par Level & Inventory Backtesting

Par Level = Lead-Time Demand + Safety Stock

Lead-Time Demand = Average Daily Demand × 3

Safety Stock = $1.645 \times \text{Daily Demand Standard Deviation} \times \sqrt{3}$

Demand statistics for the final inventory recommendation were calculated from the training period only. The test-period simulation starts inventory at the recommended par level, receives pending replenishment, deducts actual demand, records insufficient-inventory days, and orders toward par with a 3-day delivery delay.

Backtest KPI	Result
Simulated stockout days	114
Average simulated inventory	811.36 ml

Potential overstock days were identified where observed closing inventory exceeded the recommended par level. These are treated as risk signals rather than confirmed financial overstock because carrying costs and purchasing constraints were unavailable.

6. Business Implications

The solution provides a structured way to set inventory buffers by bar and brand instead of relying only on fixed manual stock levels. The results also show that intermittent demand requires careful model selection: the simple baseline remains competitive overall, while the Random Forest performs better on actual-demand days. Recommended par levels should therefore be validated against real purchasing and stockout records before production use.

7. Production Considerations & Future Improvements

- **Automation:** run daily after new POS/inventory data arrives and surface updated forecasts and reorder recommendations to managers.
- **Lead-time variability:** replace the fixed 3-day assumption with supplier-specific delivery history.
- **Demand drivers:** incorporate holidays, promotions, hotel events, weekends, and other known demand spikes.
- **Operational constraints:** model minimum order quantities, supplier capacity, delivery schedules, and storage limits.

- **Data quality:** monitor pouring waste, breakage, spillage, and inventory-count discrepancies.
- **Monitoring:** continuously track forecast error (MAE/WAPE), demand drift, stockouts, and reorder outcomes.
- **Model improvements:** evaluate Croston-style intermittent-demand methods, Exponential Smoothing, and other models using time-aware validation.